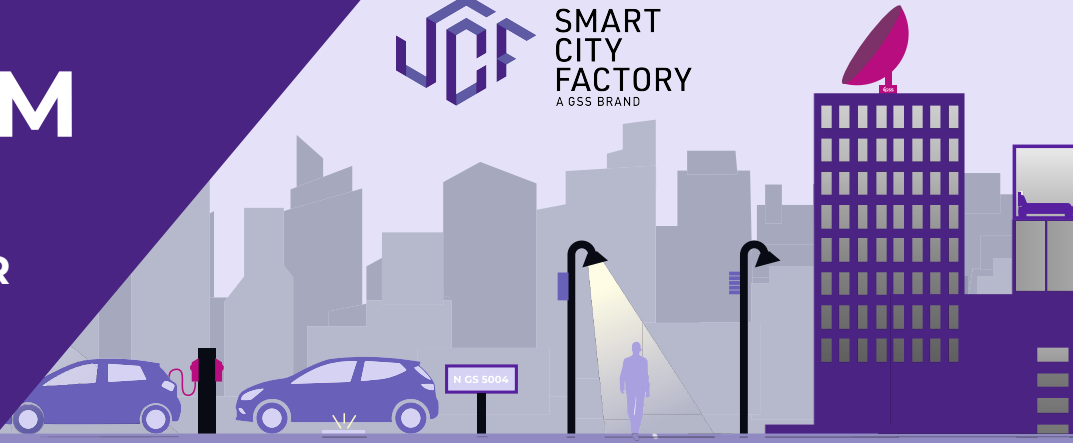


EdgeCOM

THE HARDWARE
POWERHOUSE FOR
ALL APPLICATIONS



SMART
CITY
FACTORY
A GSS BRAND



WHY CHOOSE EDGE-COMPUTING?

By processing multiple sensor data directly onsite - at the network edge (EDGE)- **EdgeCOM** enables instant analysis without relying on a continuous platform connection.

This significantly reduces latency and minimizes data traffic, as fewer data need to be transferred to remote data centers. Additionally, data security is enhanced since critical information is processed and protected locally. This makes **EdgeCOM** the ideal solution for applications that require fast responses and real-time local analysis.



APPLICATIONS



ENERGY

Optimization and control of energy flow through lighting management, consumption monitoring, and e-mobility energy management.



MOBILITY

Real-time analysis of traffic, parking spaces, and pedestrian flow using advanced sensor technology.



SENSORY

Secure data transfer of over 70 sensor variables, ranging from CO₂ levels to surface conditions.



SECURITY

GDPR-compliant monitoring for detecting threats such as aggressive behavior, broken glass, gunshots, alarm signals, and fire.

FUNCTIONS OVERVIEW

• VERSATILE

For Smart City applications, echarging stations, OEM integrations. Rugged, industrial design.

• SECURE

Secure-by-Design, Trustbooting, Self-checks.

• FLEXIBLE

Docker support for third-party applications.

• REAL-TIME

NPU for local data processing.

• ROBUST

Weatherproof and temperature resistant.

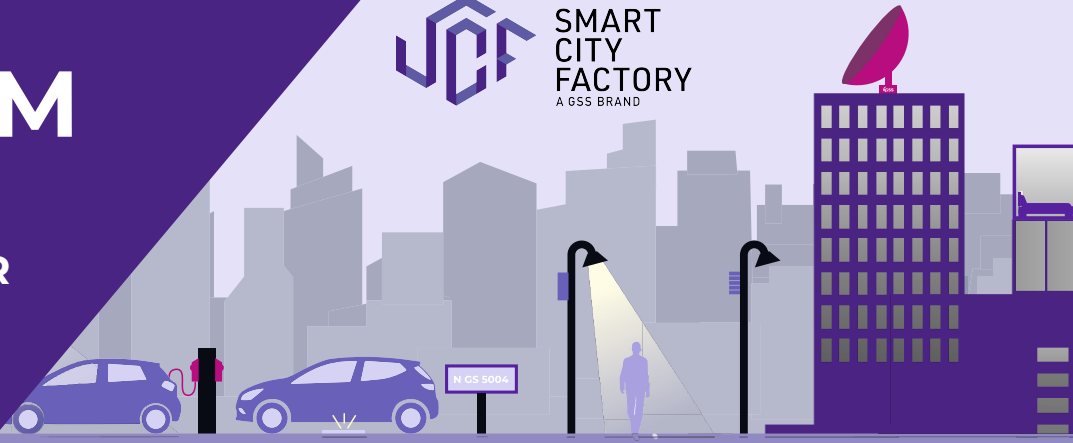
EDGECOM V1 | STAND 2025-08

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THE EDGECOM IN ACTION

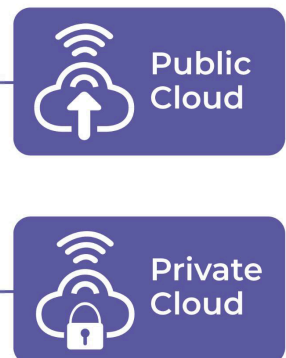
Devices



EDGECOM



Datenmanagement



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EDGECOM PROPERTIES

Processors		
CPU	ARM Cortex A53 (1,6 GHz)	
AI Accelerators	NPU 2,3 TOPS	
VPU		
Resolution	Encoding	Decoding
	1080P60	1080p60
Codec	AVC/H.264	HEVC/H.265 Main, Main 10 (up to level 5.1) (VPU G2)
	HEVC/H.265	VP9 Profile 0, 2 (VPU G2)
		VP8 (VPU G1)
		AVC/H.264 Baseline, Main, High (VPU G1)
Storage	RAM	eMMC
	1 / 2 / 4 / 8 GB LPDDR4	8 / 16 / 32 / 64 /128 GB
Connectivity		
Sensor Interfaces	1x CAN-Bus 1x RS-485 2x RJ-45 WiFi 2,4 GHz (IEEE 802.11 b/g/n) Bluetooth 2.1+EDR, BLE 5.1 Ethernet (1x WAN, 1x LAN) USB 3.0 GPIO (In/Out galvanisch getrennt)	
WAN Connectivity	Ethernet Mobile Network (4G/5G) WiFi	
GNSS	GPS, GLONASS, BDS, Galileo, QZSS	
Power supply	Input	Output (for Sensors)
	9–48 V (external PSU or PoE)	5 / 12/ 24 / 36 /48 V (Max. current: 1.8A)
Properties		
Dimensions	21 x 12 x 4,7 cm	
Weight	1,4kg	
Ambient Temperature	-40 bis 85°C	
IP Rating	IP42/IP43 (IP65/IP67 for outdoor use)	

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EDGECOM SENSORS

Environmental	
Model	Measured Value / Properties
Thies CLIMA SENSOR US	Temperature, humidity, air pressure, SO ₂ , wind (direction, speed), precipitation (rain amount/duration/intensity, hail amount/duration/intensity), solar radiation.
Thies Niederschlagsgeber 5.4041.x1.00x	Precipitation amount
Vaisala AQT410	Temperature, humidity, air pressure, NO ₂ , SO ₂ , CO, O ₃
Vaisala AQT420	Temperature, humidity, air pressure, NO ₂ , SO ₂ , CO, O ₃ , particulate matter (PM2.5 / PM10)
Vaisala AQT530	Temperature, humidity, air pressure, particulate matter (PM1 / PM2.5 / PM10), air quality (AQI), NO ₂ , CO, O ₃ , NO
Vaisala DSC211	Temperature, humidity, surface temperature, melting point temperature, surface condition*, grip level, water/snow/ice thickness
Vaisala DST111	Temperature, humidity, surface temperature, melting point temperature
Vaisala GMP252	Temperature, CO ₂
Vaisala HMP110	Temperature, humidity
Vaisala HMP110T	Temperature
Vaisala WXT 536	Temperature, humidity, air pressure, wind (direction, speed), precipitation (rain amount/duration/intensity, hail amount/duration/intensity), solar radiation
Parkin	
Ground Sensors	Features
LORA	On ground
	In ground
Optical Sensors	Detection of
	Car and bicycle parking spaces
	License plates
	Entry/Exit Tracking
Lighting control	
SCF Z-Controller	Lighting control via DALI, supporting
	Up to 64 DALI LED controllers DALI
	sensors (e.g., motion detectors)

*Values: dry, damp, wet, frosty, snow-covered, muddy, icy

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